

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence (AI) and Data Science) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Artificial Intelligence (AI) and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTAIOE505C - SOFTWARE ENGINEERING AND TESTING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Software crisis mainly refers to: A. High hardware cost B. Difficulty in managing large software systems C. Lack of programmers D. Poor operating systems
 - 2 Which model is best suited for changing requirements? A. Waterfall B. V-Model C. Agile D. Spiral
 - 3 Use cases are used to describe: A. Data flow B. System architecture C. Functional requirements D. Algorithms
 - 4 UML stands for: A. Unified Modeling Language B. Universal Model Logic C. User Model Language D. Unified Machine Language
 - 5 CASE tools are used to: A. Compile code B. Support SDLC activities C. Replace developers D. Increase errors
 - 6 Testing ensures: A. Error-free software B. Quality improvement C. Fast execution D. Low cost
 - 7 Validation is concerned with: A. Customer needs B. Code logic C. Test cases D. Documentation
 - 8 Black box testing is based on: A. Code B. Internal logic C. Specifications D. Algorithms
 - 9 Quality assurance focuses on: A. Process B. Product C. Defects D. Testing
 - 10 Bottom-up integration starts with: A. Main module B. Top module C. Low-level modules D. System
 - 11 System testing tests: A. Modules B. Integrated system C. Single function D. Code
 - 12 Adhoc testing is: A. Planned B. Structured C. Unplanned D. Automated
- Q2. Solve the following.
- A) What is SDLC? Explain each component of the SDLC. 6
 - B) Compare Between Waterfall Model and Agile Model 6
- Q3. Solve the following.
- A) Draw a Use Case Diagram For DBATU UNIVERSITY 6
 - B) Explain the software design process and design quality. 6

- Q4. Solve Any Two of the following.
- A) Differentiate between Verification and Validation. 6
 - B) Draw And Explain STLC With Their Advantages Disadvantages. 6
 - C) Compare Between White Box Testing and Black Box Testing 6
- Q5. Solve Any Two of the following.
- A) Write Short Note On Integration Testing. 6
 - B) Explain Scenario testing with system and use-case scenarios. 6
 - C) Explain Top-down integration testing with advantages and disadvantages. 6
- Q6. Solve Any Two of the following.
- A) Explain Adhoc testing with advantages. 6
 - B) Compare Between Performance Testing and Regression Testing 6
 - C) Enlist different tools used for testing object oriented software, Testing webapplications. 6

***** End *****